Watching the G1 escort policy in simulation

ECE 598JK Humanoid Escort Research | 6 November 2026 | Beginner guide

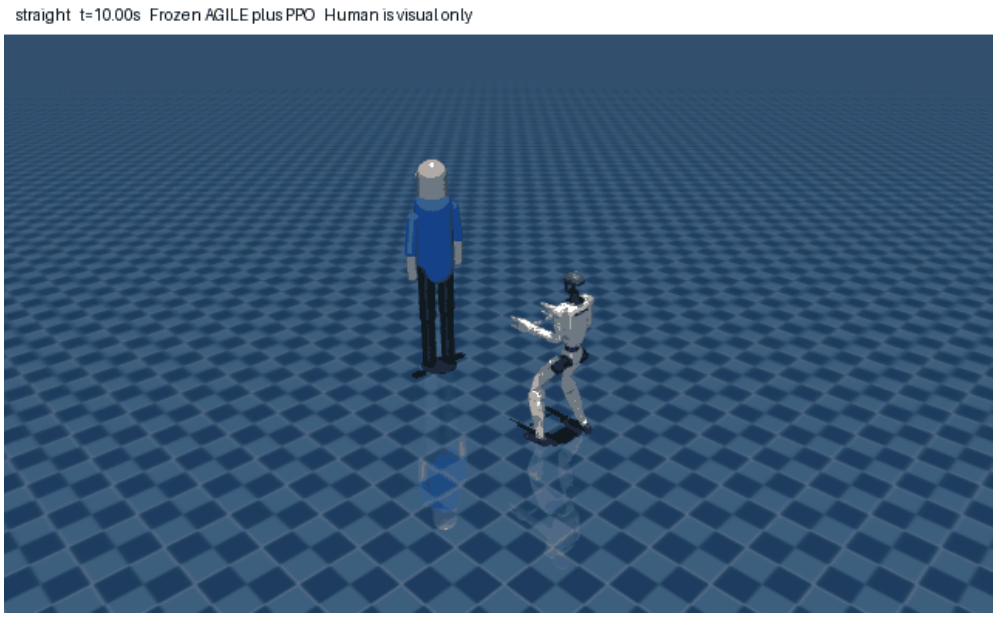
Use the existing trained policy and scenario generator to watch full-G1 escorting in three Stage 1 families. The tested host route can produce an animated clip with a human-shaped scripted client, without launching ROS or training again.

What this playback does

This player loads an existing PPO checkpoint for inference, samples one existing Stage 1 trajectory, resets its own isolated full-G1 MuJoCo environment, and runs deterministic actor actions. The G1 still walks through the frozen AGILE gait and full dynamics. No training occurs.

The blue human-shaped client is a stylized, procedurally drawn figure. Its position and heading come from the existing scenario generator. Arms and legs swing cosmetically when the current client velocity is nonzero and stop swinging when stationary. This is not a photorealistic human, a learned human motion policy or an interactive person.

The original client remains a noncolliding mocap proxy. Human geometry is added only to the rendering scene. The display uses a separate copy of the model and data, so viewer-side perturbations cannot edit the robot’s actual dynamics state. The player never publishes ROS topics, opens DDS channels or touches a running Grove simulator.



Actual offscreen playback at ten episode seconds. The G1 is governed by the trained high-level actor and frozen gait. The human figure is visual decoration around the scripted client pose.

You can use a live MuJoCo window on the host desktop or generate GIFs without a window. The host’s existing Python 3.12 venv is the verified route. Docker is not needed.

Choose the checkpoint and initialize your shell

Use a terminal on the host machine, from the repository root. The selected best model is the guarded 5k final checkpoint. The 10k final model also passed 60/60; the selector preferred the 5k model's slightly smaller mean formation error. All playback commands below default to the selected 5k model.

```
cd /home/praise/ECE598_Bodyguard/grove-g1
export
PYTHONPATH="$PWD/research/escort_rl:$PWD/research/escort_scenarios${PYTHONPATH:+:$PYTHONPATH}"
PY="$PWD/research/escort_rl/.venv/bin/python"
PLAYER="$PWD/research/escort_rl/visualization/stage1_viewer.py"
CKPT="$PWD/research/escort_rl/runs/imitation_guarded_20261106_190724_LioFor/preservation_5k/fi
nal.zip"
```

To watch the latest 10k model instead, replace only the checkpoint variable:

```
CKPT="$PWD/research/escort_rl/runs/imitation_guarded_resume_20261106_192731_IgG5HN/continuatio
n_10k/final.zip"
```

Keep the corresponding `.manifest.json` next to the ZIP. The player checks the gait and protected runtime/config hashes before running. It does not require an optimizer or RNG restore because this is inference, not training resume.

Every command uses a new output directory. It will refuse to overwrite an existing episode directory. The standalone player lives under `visualization/`, outside the trainer's source-hash globs; adding this viewer did not modify the saved training contract.

Open an interactive simulation window

After the shell setup above, run this for straight following:

```
OUT=$(mktemp -d "$PWD/research/escort_rl/runs/watch_stage1_XXXXXX")
MUJOCO_GL=glfw "$PY" "$PLAYER" \
  --checkpoint "$CKPT" \
  --family straight --seed 320000 \
  --mode viewer --client-shape human --fps 25 --speed 1 \
  --output "$OUT/straight"
printf '\nPlayback logs: %s\n' "$OUT/straight"
```

For the other families, repeat the command with a fresh `OUT` and replace `--family straight` with `--family gentle_turn` or `--family stop_start`. Each family is generated by the existing scenario wrapper; no ROS bundle or actor mailbox is created.

The private environment performs its two-second stabilization before the active scenario starts. The window follows the midpoint between the client and robot. Ordinary MuJoCo camera controls let you rotate and zoom. Closing the window stops that playback and records `viewer_closed`, not success. Let the episode finish to obtain its task outcome. Use `--speed 0.5` for slower wall-clock playback; the physics timestep, actor timing and simulated route remain unchanged.

`--fps 25` draws every 0.04 simulated seconds, while PPO still acts every 0.2 s and physics still steps every 0.002 s. Rendering interpolated-looking motion does not add policy decisions. Supported frame rates are 5, 10, 20, 25 and 50.

Validation status: offscreen EGL rendering and complete recorded episodes were tested on this host. The desktop interactive-window path is provided but was not opened automatically. If the window cannot connect to a display, use the GIF route below.

Record all three scenarios without a display

This is the fully tested route. It uses existing EGL support and Pillow; it needs no ffmpeg, ROS, Docker, browser server or package installation.

```
OUT=$(mktemp -d "$PWD/research/escort_rl/runs/stage1_gifs_XXXXXX")
for FAMILY in straight gentle_turn stop_start; do
  MUJOCO_GL=egl "$PY" "$PLAYER" \
    --checkpoint "$CKPT" \
    --family "$FAMILY" --seed 320000 \
    --mode gif --client-shape human --fps 10 \
    --width 640 --height 400 \
    --output "$OUT/$FAMILY" || break
done
printf '\nGIFs and logs: %s\n' "$OUT"
```

Open each `episode.gif` in your browser or image viewer. For example:

```
xdg-open "$OUT/straight/episode.gif"
```

A typical clip uses roughly 38–41 simulated seconds and 385–407 frames at 10 fps. These examples are around 45 MB each. GIF export runs as fast as compute/rendering allows; its frame durations produce normal-speed playback. The `--speed` option is relevant to the interactive window, not GIF timing.

For a quick two-second graphics check, add `--max-seconds 2` to a command with a fresh output path. The result will be labelled `preview_limit` and is not a completed task. Use `--client-shape capsule` to inspect the original simple client appearance.

What to watch in each family

Family argument	What the client does	What to inspect in the G1
straight	Smooth acceleration along a straight route, then endpoint deceleration.	Maintains the 1.5 m trailing offset, corrects small initial errors and settles at arrival.
gentle_turn	Follows a smooth curved path, with heading tangent to motion.	Turns with the client and adjusts lateral motion while keeping the offset in client axes.
stop_start	Decelerates, pauses for 1–3 seconds, resumes, then stops at the endpoint.	Avoids walking through the client during the pause and resumes without losing formation.

All families contain one client and zero pedestrians. Peak-speed settings are sampled from 0.15–0.25 m/s; route length is 3–5 m. Turns use curvature magnitude 0.06–0.16 per metre and a 0.06 rad/s turn-rate limit; scenario acceleration limit is 0.16 m/s². Heading, position, formation error, turn direction and stop timing are seeded random variables.

The example seed 320000 is a development seed accepted for all three families. Reusing it across families gives a reproducible demonstration, not three identical trajectories. Use another nonnegative seed divisible by four to explore more routes. Reserved final-test speed/length combinations are rejected explicitly; choose a different development seed rather than overriding the check.

The trailing point rotates with client heading. A robot standing directly south of the client in world coordinates may be on the wrong side after a turn. Judge formation in the client's heading frame rather than by a fixed screen direction.

Read the outcome instead of judging only the animation

Each episode folder contains the following:

File	Purpose
episode.gif	Rendered clip when mode is gif.
summary.json	Outcome, seed, family, model hash, PPO timestep count and episode metrics.
trace.jsonl	Every high-level observation, predicted action, reward, ending flags and diagnostic info.
scenario_evaluator_metadata.json	Resolved route parameters for inspection, separate from actor input.

A complete success has `status: success`, `full_episode_completed: true` and `new_training_updates: 0`. In `episode_metrics`, check formation mean/p95 and reward component totals. Per-step `info.evaluation` includes signed formation errors, progress and stopping-hold diagnostics. The trace also distinguishes requested commands from measured body velocity.

After the three-GIF command, print a compact result table:

```
"$PY" - "$OUT" <<'CHECK'
import json, sys
from pathlib import Path
for name in ("straight", "gentle_turn", "stop_start"):
    r = json.loads((Path(sys.argv[1])/name/"summary.json").read_text())
    m = r.get("episode_metrics", {})
    print(name, r["status"], "formation m", m.get("formation_mean_m"),
          "completed", r["full_episode_completed"])
CHECK
```

These three demonstrations are visualization checks, not a replacement for sixty-case validation. The previously completed strict suite passed 60/60 at 5k and 10k, but repeated use of that suite and one training seed limits broader claims.

Inspect or export the scenario without loading the robot

If you want the client route plot and metadata before a simulation, reuse the existing scenario exporter. Its trajectory file is for evaluation and visualization only; do not feed it into the actor.

```
SCENARIOS=$(mktemp -d "$PWD/research/escort_rl/runs/stage1_routes_XXXXXX")
for FAMILY in straight gentle_turn stop_start; do
  "$PY" research/escort_rl/tools/stage1_scenarios.py     --config
research/escort_rl/configs/stage1.yaml     --family "$FAMILY" --seed 320000     --output
"$SCENARIOS/$FAMILY" || break
done
```

This exports client trajectory CSV, metadata, validation results, an XY plot and time-series plots. Client pose is still sampled at current simulation time during actual playback; the renderer does not replace the generator or move the G1 along a prerecorded path.

Troubleshooting and Docker boundaries

A desktop window does not open

Check `echo "$DISPLAY"`. The window command needs an accessible Linux display and GLFW. An SSH terminal may not have one. Use `--mode gif` with `MUJOCO_GL=egl` instead. Do not launch or reset the Grove ROS graph to fix an isolated visualization issue.

EGL fails on a different machine

This host's EGL route was tested. Driver/device access can differ elsewhere. If available, try `MUJOCO_GL=osmesa` as an alternative offscreen backend. That fallback was not tested here and may require libraries. Stop and inspect the error; do not install system packages automatically.

Imports fail

Run the shell setup again and use the exact venv Python. Missing `escort_scenarios` usually means its source directory is absent from PYTHONPATH. Missing Torch, MuJoCo or SB3 usually means the wrong interpreter. Do not use system Python as an unverified substitute.

A checkpoint or hash check fails

Keep its manifest alongside the ZIP and inspect the reported mismatch. Loading the wrong reward/runtime version can make a visual demonstration misleading. Do not bypass identity checks or edit the old manifest.

Can this run in Docker

The host commands above are the recommended complete route because the pinned host venv and EGL are verified. Do not assume a host venv is portable into a container. The container would need the same Python packages and asset paths, the repository and checkpoint mounted, and working GL/device or desktop access. Those conditions were not verified, so there is no claimed turnkey Docker command in this guide. No Docker lifecycle or dependency-install command is necessary for the tested host route.

The player is not an adapter for an existing live simulator. It creates its own MuJoCo instance. Watching the policy in the Grove ROS world would be a separate integration milestone requiring verified command ownership and the collision-monitor chain.

What was actually verified for this guide

Four new tests check finite human geometry, stopping of cosmetic swing, and identical physics states, controls, observations and rewards with visualization callbacks enabled for all three families. A deliberate perturbation of the display copy does not change the dynamics instance.

Three complete EGL-rendered demonstrations with the selected 5k policy, seed 320000, produced:

Family	Outcome	Active duration	Mean formation error
Straight	Success	38.4 s	0.0602 m
Gentle turn	Success	38.4 s	0.0664 m
Stop and start	Success	40.6 s	0.0510 m

The full regression suite passed 113 tests with zero failures or skips. Its log and the sample clips are stored under `research/escort_rl/runs/stage1_visualization_20261106/`. Existing gait weights, controllers, physics, rewards and observation definitions were not modified. No PPO updates, live ROS interactions or Docker lifecycle commands occurred.

Open the already-generated clips from this HTML guide:

- Straight following demonstration
- Gentle turning demonstration
- Stop and start demonstration

The human is decorative and noncolliding. The policy still receives numerical proxy state. These clips establish visualization of actual isolated full-G1 dynamics, not perception, responsive human behavior, live-stack parity or physical-person safety.